

Optimal Conditions for Boiling Water

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MTH 547 - Design of Experiments



Objective of the Study

- Determine how **initial water temperature** (70°, 90°, 110°) and **salt concentration** (0, 1, 3 tsp) affect **boiling time**.

Motivation

- A common belief is that hotter water boils faster.
- Some also claim that adding salt or starting with colder water affects boiling time.
- This study tests these claims using a controlled experiment.

Data

Block 1 - Electric Coil									
Temperature (°F)									
Salt Level (tsp)	70			90			110		
0	245	219	223	199	188	186	174	180	162
1	259	220	224	189	194	215	188	178	172
3	217	208	230	212	200	197	178	181	183

Block 2 - Electric Stovetop (Glass)									
Temperature (°F)									
Salt Level (tsp)	70			90			110		
0	262	263	242	223	231	200	201	199	189
1	249	250	255	222	208	230	201	207	203
3	260	246	250	220	226	244	197	204	197

Experimental Setup

- The team used two identical pots and thermometers, with two cups of water in each trial.
- Stovetop type (electric coil vs glass-top) was treated as a blocking factor.
- The order of treatment combinations was randomized to reduce bias.
- Each treatment combination was replicated 3 times.
- Response: Time (seconds) to reach 210°F.



Results

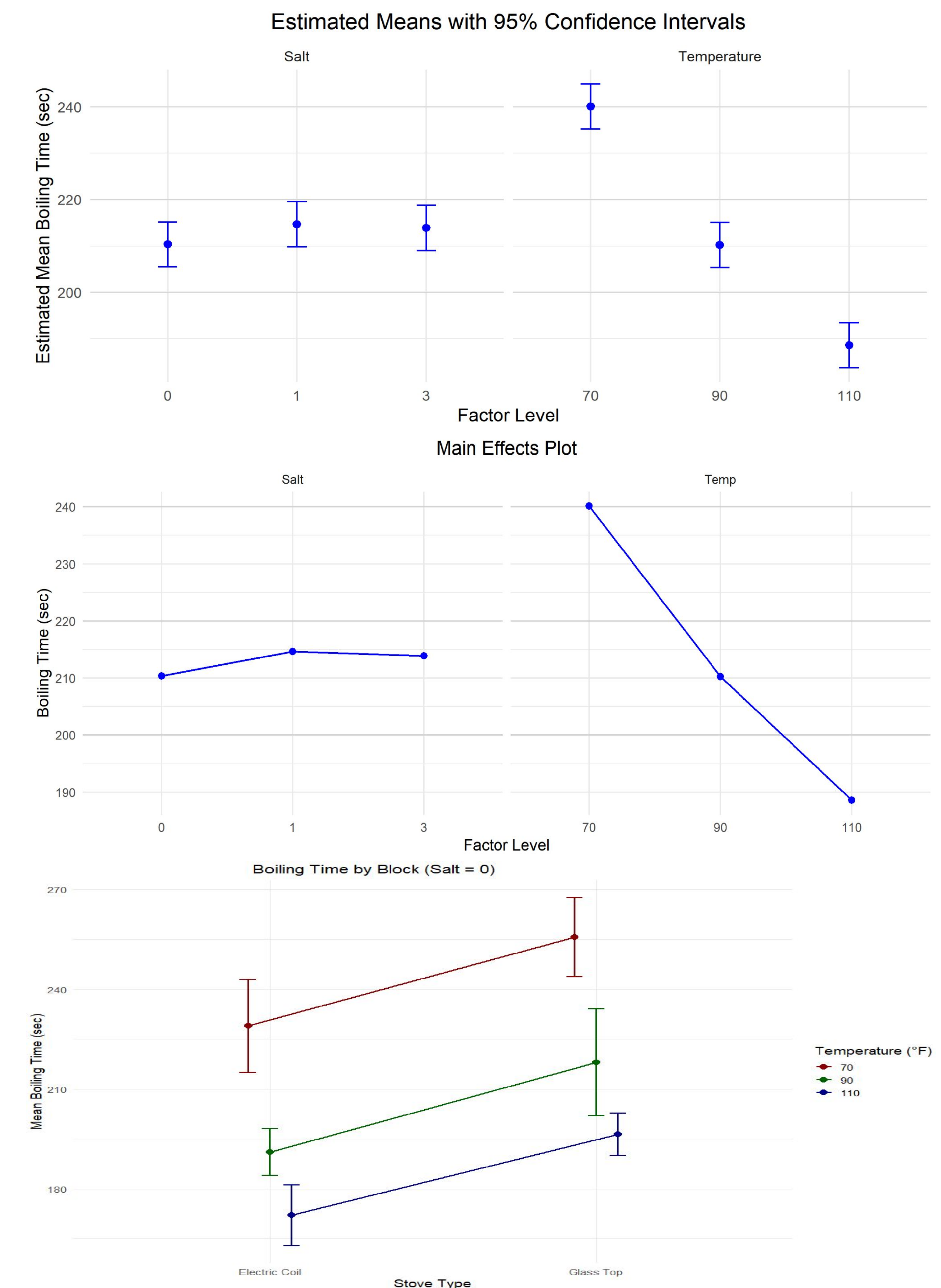
- Interaction Effect:** Testing indicated that interaction effects were not statistically significant; therefore, the interaction term was removed from the final ANOVA model.

- Final model:**
 $\text{Boiling Time} = \mu + \text{Block} + \text{Salt} + \text{Temperature} + \epsilon$

ANOVA Table

	Df	Sum Sq	Mean Sq	F-Value	Pr (>F)
Block	1	8018	8018	76.310	1.74e-11
Salt	2	192	96	0.914	0.408
Temperature	2	24125	12062	114.803	< 2e-16
Residuals	48	5043	105		

- Main Effects:** The refined model shows that temperature has a statistically significant effect on boiling time ($p < 0.05$), while salt concentration is not significant over the tested range.
- Stove Type Comparison:** The electric coil stove consistently produced faster boiling times compared to the glass.
- Temperature Trend:** Estimated mean boiling time decreases as initial temperature increases.
- Block Effect:** An electric coil stove heats and boils water faster than a glass-top stove.
- Conditions for constant variance and normality have been checked with no concerns of violations for either.



Discussion & Conclusions

- Salt concentration was not found to have a significant effect on boiling time.
- Electric coil stovetops produced faster boiling times than glass stovetops.
- The optimal condition for fastest boiling occurs with high starting temperature on an electric coil stovetop.

Limitations

- A gas stovetop was not available, so only two block levels were tested.
- Only tested temperature and salt levels.